

Division of integers

The same sign rules as multiplication — and the one thing that is never allowed.

LESSON 31–32

2 × 40 MIN

MATEMATIKA 7, §13

S8 1.2

LESSON CLOCK

15'

25'

25'

20'

5'

The sign rules	15 min
Division by zero	25 min
Order of operations	25 min
When is it an integer?	20 min
Homework	5 min

LEARNING OBJECTIVES

- Divide integers using the sign rules.
- Explain why division by zero is undefined.
- Decide whether a quotient of two integers is an integer.
- Use the order of operations with signed numbers.

TEXTBOOK

National	pp. 83–88
Cambridge	Stage 8, pp. 6–10
Workbook	Exercise 1.2

01

Explanation

The sign rules

Division is the reverse of multiplication, so the signs behave in exactly the same way.

SIGNS	QUOTIENT	EXAMPLE
$(+) \div (+)$	+	$20 \div 5 = 4$
$(+) \div (-)$	–	$20 \div (-5) = -4$
$(-) \div (+)$	–	$(-20) \div 5 = -4$
$(-) \div (-)$	+	$(-20) \div (-5) = 4$

Why. $(-20) \div (-5) = 4$ because $4 \times (-5) = -20$. Every division is checked by the multiplication that undoes it.

CHECK EVERY DIVISION BY MULTIPLYING BACK

It takes two seconds and catches both a wrong size and a wrong sign. It is the only check that needs nothing but the answer you already have.

Division by zero

$a \div 0$ is undefined for every a

Why. $20 \div 0$ would be a number q with $q \times 0 = 20$ — but every product with zero is zero, so no such q exists.

EXPRESSION	VALUE
$0 \div 7$	0
$7 \div 0$	undefined
$0 \div 0$	undefined

$0 \div 7$ AND $7 \div 0$ ARE COMPLETELY DIFFERENT

The first is 0 ; the second does not exist. Zero on the top is fine, zero on the bottom never is — a rule that will matter in every algebra lesson from now on.

Order of operations

ORDER	OPERATION
1	brackets
2	powers
3	multiplication and division, left to right
4	addition and subtraction, left to right

Example. $-12 \div 4 + (-3) \times 2$. Divide and multiply first: $-3 + (-6)$. Then add: -9 .

LEFT TO RIGHT WITHIN THE SAME LEVEL

$24 \div 4 \times 2$ is 12 , not 3 : division and multiplication have equal rank, so they are done in the order written.

When is it an integer?

A quotient of integers is an integer only when the divisor is a factor of the dividend.

DIVISION	INTEGER?	VALUE
$(-36) \div 9$	yes	-4
$(-36) \div (-12)$	yes	3
$(-36) \div 5$	no	-7.2
$7 \div (-2)$	no	-3.5

This is why $\mathbb{Z}$ is not closed under division, and why the next chapter introduces the rational numbers.

THE SIGN HAS NOTHING TO DO WITH IT

Whether the answer is a whole number depends only on the moduli. The sign is decided separately, and never affects divisibility.

02

Worked examples

EXAMPLE 1 Compute $(-20) \div (-5)$, $20 \div (-5)$ and $(-36) \div 9$.

1

Alike signs: positive. $20 \div 5 = 4$.

2

Different signs: negative. -4 .

3

Different signs: $36 \div 9 = 4$, so -4 .

4

Check: $(-4) \times 9 = -36 \checkmark$

ANSWER

$4, -4, -4$

EXAMPLE 2 Compute $-12 \div 4 + (-3) \times 2$.

- ① Division and multiplication first.
- ② $-12 \div 4 = -3$ and $(-3) \times 2 = -6$.
- ③ $-3 + (-6)$
- ④ $= -9$

ANSWER

 -9 **EXAMPLE 3** Explain why $7 \div 0$ has no value, while $0 \div 7 = 0$.

- ① $7 \div 0 = q$ would mean $q \times 0 = 7$.
- ② Every product with zero is zero, never 7.
So no q exists.
- ③ $0 \div 7 = q$ means $q \times 7 = 0$.
- ④ $q = 0$ works — and only that.

ANSWER

Undefined; and 0

03 Interactive model

Ask how many times zero fits into seven; the class runs out of patience before the answer, which is exactly the point.

This model could not start: den is not a function

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Division	Bo‘lish	Деление
Quotient	Bo‘linma	Частное
Dividend	Bo‘linuvchi	Делимое
Divisor	Bo‘luvchi	Делитель
Undefined	Aniqlanmagan	Не определено
Order of operations	Amallar tartibi	Порядок действий
Exactly divisible	Butun bo‘linadi	Делится нацело
Remainder	Qoldiq	Остаток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$(-20) \div (-5)$ equals:

$20 \div (-5)$ equals:

$0 \div 7$ equals:

$7 \div 0$ is:

0

7

undefined

∞

$24 \div 4 \times 2$ equals:

3

12

48

192

$(-36) \div 5$ is:

an integer

not an integer

zero

undefined

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $20 \div 5$

ANSWER 4

2. $20 \div (-5)$

ANSWER -4

3. $(-20) \div 5$

ANSWER -4

4. $(-20) \div (-5)$

ANSWER 4

5. $0 \div 9$

ANSWER 0

6. $9 \div 0$

ANSWER Undefined

7. $(-36) \div 9$

ANSWER -4

Medium · the standard question

7 PROBLEMS

1. $(-36) \div (-12)$

ANSWER 3

2. $(-45) \div 9$

ANSWER -5

3. $-12 \div 4 + (-3) \times 2$

ANSWER -9

4. $24 \div 4 \times 2$

ANSWER 12

5. $(-18) \div (-3) - 4$

ANSWER 2

6. $(-2)^3 \div 4$

ANSWER -2

7. Is $(-36) \div 5$ an integer?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. $(-48) \div (-6) \div (-2)$

ANSWER -4

2. $(-3)^2 \times 2 - 30 \div (-5)$

ANSWER 24

3. $(-60) \div (2 - 5)$

ANSWER 20

4. Find x : $\frac{x}{-4} = 7$

ANSWER -28

5. Find x : $\frac{-36}{x} = -9$

ANSWER 4

6. For which integers n is $\frac{12}{n}$ an integer?

ANSWER $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$

7. Why is $0 \div 0$ undefined?

ANSWER Every number would satisfy it

07 Homework**Homework — 5 tasks**

Check every division by multiplying the answer back.

1. Compute $(-42) \div 7$, $42 \div (-7)$ and $(-42) \div (-7)$.

2. Compute $-20 \div 5 + (-4) \times 3$.

3. Compute $(-2)^4 \div (-8)$.

4. Explain why $5 \div 0$ is undefined.

5. List all integers n for which $\frac{18}{n}$ is an integer.